Inequality and the US Macroeconomy

Lag-structure evidence from annual data, 1948–2025 (n = 78 complete cases)

Data and method

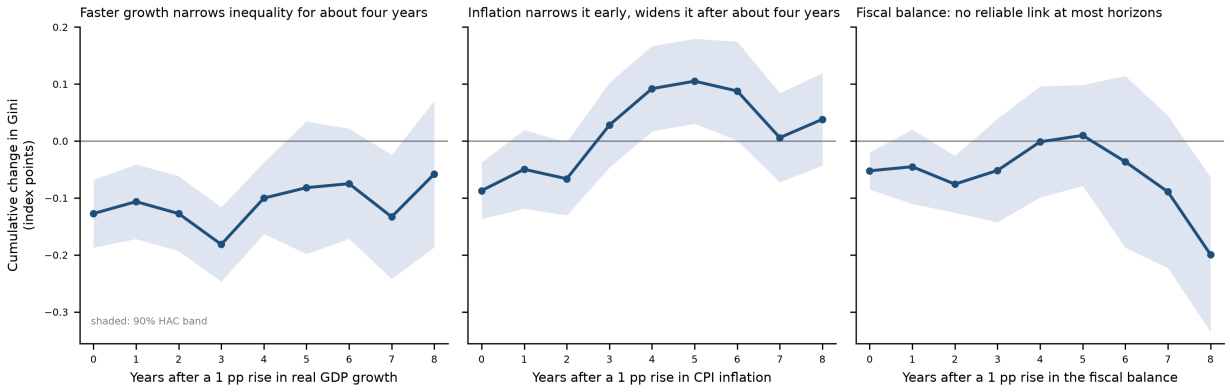
Six annual US series were merged into a single panel. Inequality is the Census Gini index distributed by FRED as *GINIALLRF* (1947–2025). The macro block is real GDP growth (A191RL1A225NBEA), CPI inflation (CPIAUCSL), the federal surplus or deficit as a share of GDP (FYFSGDA188S) and the unemployment rate (UNRATE). Monthly series were aggregated to calendar-year averages using complete years only, so 2026 is excluded rather than averaged over nine months. Inflation is carried in two forms, annual-average and December-over-December; the two differ materially in individual years (4.68 against 7.17 percent in 2021), so the baseline choice of annual-average is explicit rather than inherited from a frequency-conversion default.

Dynamics are estimated by Jordà local projections: the cumulative change in the Gini from the year before the observation to horizon h is regressed on the macro variable in year t, controlling for two lags of all five variables and of the Gini change itself. Horizons run 0 to 8 years, standard errors are Newey–West with h+1 lags, and the estimation sample falls from 76 observations at impact to 68 at the longest horizon. Coefficients read as Gini index points per 1 percentage point.

Findings

Macro variable	Impact (h=0)	Largest effect	90% band at largest	Horizons with band off zero
Real GDP growth	-0.127	-0.181 at 3 yr	[-0.247, -0.116]	0, 1, 2, 3, 4, 7
CPI inflation (annual average)	-0.087	+0.105 at 5 yr	[+0.031, +0.179]	0, 2, 4, 5, 6
Federal fiscal balance (% GDP)	-0.052	-0.199 at 8 yr	[-0.335, -0.064]	0, 2, 8

- **Growth is the one clear result.** A 1 pp higher real growth rate predicts a Gini 0.127 points lower on impact, deepening to -0.181 at 3 years [-0.247, -0.116], then fading. The band excludes zero at horizons 0, 1, 2, 3, 4, 7.
- **Inflation reverses sign.** Negative on impact (-0.087), still negative at two years, crossing into positive territory from year four and growing to a peak of +0.105 at 5 years [+0.031, +0.179]. A single contemporaneous regression would net these two phases against each other and report approximately nothing.
- **The fiscal balance mostly is not there.** Negative at impact (-0.052) and at two years, then a band straddling zero. The large -0.199 at 8 years sits on the widest band in the figure and the smallest subsample, and should not carry weight.
- **Magnitudes are small.** The Gini rose 8.9 points across the sample (37.1 to 46.0). The largest growth effect is 2.0 percent of that rise, so these are cyclical modulations around the trend, not an account of the trend itself.



Cumulative Gini response to each macro variable. Shaded bands are 90 percent Newey–West intervals. Responses are predictive, not identified causal effects.

What these estimates do and do not establish

Identification. The regressor is the realised macro variable, not an identified shock. These are predictive responses conditional on two lags of the system, so reverse causality running from inequality to growth is unaddressed, as is any third factor moving both. The literature's standard remedies are narrative or sign-restricted shocks for monetary and fiscal policy, or an instrument for growth; none is applied here.

The inequality concept. GINIALLRF appears to be the Census *family* Gini rather than the household series of Table H-4: the identifier suffix and the 1947 start both point that way, and household data begin only in 1967 and sit several points higher. The downloaded file carries no series title, so this identification is inferred and should be confirmed against FRED's documentation before the number appears in a chapter. Families exclude single-person households, which is not a neutral omission here: the growth in single-person households is itself part of the measured rise in inequality.

Sample and specification. Annual data over 78 years supports two lags and little more. Horizon-8 estimates rest on 67 observations. The lag length, the horizon ceiling and the control set were not selected by criterion; they are conventional choices, and the growth result should be checked against alternatives before it is leaned on. Stationarity was not formally assessed: the dependent variable is a cumulative change, which handles the Gini's persistence, but the regressors enter in levels.

Suggested next steps

1. **Vary the inequality concept.** Rerun with the Census household Gini (Table H-4, 1967 onward) and the WID top 1 percent income share (1913 onward). If the growth and inflation patterns hold across a dispersion measure and a concentration measure, the result is considerably stronger than it is now; if they do not, the finding is about family income composition rather than inequality in general. The WID series also roughly doubles the sample, which would let the lag structure be estimated rather than assumed.

2. **Test the inflation-timing choice.** Re-estimate using December-over-December inflation, already present in the panel. The mid-horizon sign flip is the paper's most interesting feature and also the one most exposed to how inflation is dated against annual income.

3. **Move toward identification.** Substitute identified monetary and fiscal shock series for the realised variables. This is the step that converts the growth coefficient from a stylised fact into a causal claim, and it is where a referee will look first.

4. **Extend to the cross-country panel.** The original design: World Bank WDI Gini with IMF WEO growth, inflation and fiscal balances, estimated with country and year fixed effects, then the same local projections within panel. Cross-country variation also allows the institutional and redistributive differences that a single-country time series cannot speak to.

5. **Resolve data access before scaling up.** Every source needed for the cross-country phase is blocked by the organisation's network allowlist, so each file currently arrives by manual download. Requesting that an administrator allowlist fred.stlouisfed.org, www2.census.gov, wid.world and api.worldbank.org would make the whole pipeline a rerunnable fetch rather than a file shuffle, which matters once the panel spans many countries and vintages.

Reproducibility

Saved alongside this note: *us_macro_annual.csv* (the merged annual panel), *lp_gini_response.csv* (all 27 coefficients with standard errors, bounds, t-statistics and per-horizon n) and *gini_local_projections.png*. Each carries the code that produced it, so the aggregation conventions above are recoverable rather than described. Source files were downloaded from FRED on 15 October 2026.